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A stealthy and robust backdoor attack via frequency domain transform

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Abstract
Deep learning models are vulnerable to backdoor attacks, where an adversary aims to inject a hidden backdoor into the deep learning models, such that the victim models perform well on clean data but output predefined wrong results on data containing specific triggers (e.g., a pattern, or a specific accessory). While existing attack methods are effective, they are commonly not stealthy and robust, i.e., the backdoor triggers are unnatural and easily detected, and they are hard to resist data augmentation operations. To address these issues, in this paper, we explore new types of attack methods that significantly improve the stealthiness and robustness of backdoor attacks. Specifically, inspired by digital watermarking techniques, we propose two backdoor trigger injection algorithms based on discrete Fourier transform and discrete cosine transform. These algorithms select the frequency domain instead of the spatial domain for trigger injection, ensuring the stealthiness. Besides they divide the original data into multiple data blocks for multiple injections of triggers to improve the robustness. We experimentally evaluated the proposed methods on GTSRB and CIFAR10 datasets, and the results demonstrate that our methods remarkably improve the stealthiness and robustness of backdoor attacks without compromising effectiveness. For example, on GTSRB, compared with the Badnets and Blend, our methods generate more natural poisoned data, and improve at least 80.99%, 68.09%, 25.49%, and 63.31% in random horizontal flip, random vertical flip, random cropping (padding=2), and random cropping (padding=4).

Keywords Backdoor attacks · Frequency domain transform · Effectiveness · Stealthiness · Robustness

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1 Introduction

In recent years, deep learning has been overwhelmingly successful in many domains [1–5], such as computer vision, speech recognition, and natural language processing. But deep learning relies on massive training data and high-performance computing resources [6, 7], such as GPU and RAM. These resources result in high computational costs, severely limiting the study and use of deep learning by individuals and most groups. To address this limitation, they outsourced the training process to the cloud server providers, i.e., Machine Learning as a Service (MLaaS) [8]. MLaaS is an emerging cloud computing service that provides users with fast machine learning tools and low-cost high-performance computing resources, which significantly improves the usability of deep learning. However, the opacity of the MLaaS poses new security threats to deep learning models [9–13], such as backdoor attacks.

Gu et al. first discovered backdoor attacks on deep learning models in 2019 [14]. They add specific triggers to a small amount of training data and assign the wrong labels to them. Then these data are used to train the model. The trained model would output predefined wrong results on data containing triggers but perform well on clean data. The discovery of backdoor attacks illustrates the fragility of deep learning and heightens concerns about deep learning security in academia and industry. To explore the internal mechanism of backdoor attacks, researchers have conducted in-depth studies on backdoor attacks and proposed many attack methods. However, these attack methods face many challenges in the real world. For example, the backdoor triggers are commonly unnatural and easily detected, as shown in Figure 1. And they are prone to failure after data augmentation operations such as flip and cropping. Hence, the stealthy and robust backdoor attacks are still open issues.

Stealthiness and robustness are also the basic requirements of digital watermarking [15, 16]. Digital watermarking is an information hiding technology that could invisibly embed identification information into the picture, video, audio, text, and other carrier data. Since the watermark information is below the minimum granularity that the human sensory system can bear, it can be used to prove the source of the data and protect its integrity. Inspired by digital watermarking techniques, we fully consider the effectiveness, stealthiness, and robustness of backdoor attacks, and propose new types of attack methods in this paper. To mount the attack, the adversary only adds imperceptible perturbations to a small amount of training data, and does not manipulate the training process and parameters of models. Specifically, we design two backdoor trigger injection algorithms based on discrete Fourier transform (DFT) and discrete cosine transform (DCT). These algorithms first transform the image from the spatial domain to the frequency domain, then inject triggers into the frequency domain of the image, and finally transform the image from the frequency domain back to the



Figure 1 Examples of poisoned data (Left: Badnets. [14] Middle: Blend. [17] Right: Trojan. [18])

spatial domain. Compared with the existing backdoor attack methods, our methods select the frequency domain instead of the spatial domain for trigger injection to ensure the stealthiness. And they divide the original data into multiple data blocks for multiple injections of triggers to improve the robustness.

The main contributions of this paper are as follows:

1. We discuss the challenges faced by backdoor attacks on deep learning models. Specifically, existing attack methods have poor stealthiness and robustness, and are difficult to apply in the real world.
2. We propose two backdoor trigger injection algorithms based on DFT and DCT. They transform the image from the spatial domain to the frequency domain and inject the triggers into the frequency domain multiple times.
3. We experimentally evaluated the proposed methods on GTSRB and CIFAR10 datasets, and the results show that our methods significantly improve the stealthiness and robustness of backdoor attacks without compromising effectiveness.

The subsequent sections of the paper are structured as follows: Section 2 introduces the related work. The methods are described in detail in Section 3. Section 4 provides evaluation results of the methods. At last, the paper is concluded in Section 5.

2 Related work

2.1 Backdoor attack

The backdoor attack is an emerging and rapidly growing research area [19]. Existing attack methods can be categorized into visible backdoor attacks and invisible backdoor attacks according to whether the trigger is visible.

2.1.1 Visible backdoor attacks

Gu et al. first defined backdoor attacks on deep neural networks and proposed a backdoor attack method *Badnets* [14]. It adds the specific trigger to clean data and assigns wrong labels to them. Then these data are used to train the model. After training, the model will output predefined wrong results on data containing triggers but perform well on clean data. In the black-box scenario, Liu et al. proposed a backdoor attack method [18]. They selected some data from the public dataset and optimized them to match the output of the target model. Then they generated poisoned data by adding triggers to the optimized data. Bagdasaryan et al. studied semantic backdoor attacks [20]. They use semantics as backdoor triggers instead of a specific pattern, such as some type of racing stripes on a car.

2.1.2 Invisible backdoor attacks

Chen et al. first considered invisible backdoor attacks and proposed a blending-based trigger injection method [17]. Its basic principle is $\Phi(t, x) = \alpha \cdot t + (1 - \alpha) \cdot x$. And t is the trigger, x is the clean data, α is the balance parameter that controls the weight of the trigger and the clean data. Compared with *Badnets*, this method uses global Gaussian noise instead of the pixels as a backdoor trigger. Turner et al. proposed the clean-label backdoor attack [21]. They conduct untargeted adversarial attacks on clean data to mitigate the effects of robust

features, which realizes the backdoor attacks without modifying the original labels. Saha et al. optimized the similarity between clean data and poisoned data in the feature space to mount the clean-label backdoor attack [22]. Quiring et al. used the scaling attack that scales the picture to a specific resolution to inject backdoor triggers [23]. Zhong et al. adopted the idea of universal adversarial perturbation to generate backdoor triggers [24, 25]. They used the l_2 norm to constrain the size of the trigger. Liu et al. explored reflection phenomena in nature and used them as backdoor triggers [26]. Li et al. proposed a backdoor attack method using autoencoders that takes the original clean image as input and injects sample-specific triggers via data reconstruction [27].

2.2 Backdoor defense

To mitigate backdoor attacks, researchers have studied the defense methods against backdoor attacks [19, 28]. Existing defense methods can be divided into input-preprocessing-based defense methods, model-diagnosis-based defense methods, model-reconstruction-based defense methods, and other defense methods.

2.2.1 Input-preprocessing-based defense methods

Input preprocessing refers to eliminating triggers in the input data before model inference. Liu et al. used the autoencoder as an input preprocessor to filter the input data before model inference, deactivating the triggers [29]. Another input-preprocessing-based defense method is proposed by Doan et al. [30]. It covers the part of the data that contributes the most to the prediction, and reconstructs the covered part via image restoration method. Li et al. found that changing the appearance or location of triggers drastically reduced the effectiveness of backdoor attacks [31]. Hence, they employ image processing operations to build input preprocessors.

2.2.2 Model-diagnosis-based defense methods

The basic principle of model diagnosis is to determine whether the model contains the backdoor via some technical means. Wang et al. proposed a model diagnosis method based on trigger synthesis [32]. They generate potential backdoor triggers for each class and use anomaly detection to determine whether the model contains a backdoor. Inspired by universal adversarial perturbations, Kolouri et al. introduced universal litmus patterns that are essentially optimized images [33]. They determine whether the model contains a backdoor based on its output on the universal litmus pattern. Huang et al. found that the explainable information on clean and poisoned data has different characteristics [34], including sparseness, smoothness, and persistence. They used these features to build detectors to discriminate between clean and poisoned data.

2.2.3 Model-reconstruction-based defense methods

Model reconstruction intends to modify the structure or parameters of the model to eliminate the backdoor. Due to the catastrophic forgetting property of deep learning models, Liu et

al. retrained the model using the clean dataset [29]. Gu et al. found that the clean data does not activate neurons associated with the backdoor [14], so Liu et al. removed the neurons associated with the backdoor using neural network pruning [35].

2.2.4 Other defense methods

Except for the above defense methods, Du et al. proposed a poison-suppression-based defense method [36]. It applies differential privacy techniques to model training [37, 38], which reduces the effectiveness of the backdoor trigger and increases the difficulty of backdoor injection. Chen et al. believe there is a difference in the feature distribution between poisoned and clean data, so they used neuron activations clustering to detect abnormal data containing backdoor triggers [39]. Gao et al. superimposed the suspect data with other data [40], then input them into the target model, and judged whether the data contains triggers through the randomness of the prediction results. The less random the output is, the more likely the model contains the backdoor.

2.3 Digital watermarking

Digital watermarking is an effective method to realize anti-counterfeiting traceability and copyright protection. Existing digital watermarking methods can be divided into spatial domain and frequency domain digital watermarking.

2.3.1 Spatial domain digital watermarking

The spatial domain digital watermarking methods directly modify the pixel values to realize the watermark embedding. To reduce the impact of watermark embedding on data quality, Van Schyndel et al. embedded the watermark into the least significant bits of pixel values [41]. This method can obtain a larger watermark embedding space, but the watermark is fragile and easily damaged. Bender et al. proposed a statistical-based digital watermarking method [42]. That selects n pairs of pixels and embeds the watermark via increasing the brightness of one pixel and reducing the brightness of another pixel. This method has good stealthiness, but it is only suitable for images containing a lot of textures.

2.3.2 Frequency domain digital watermarking

Frequency domain digital watermarking adds the watermark in the frequency domain space of the image. Compared with the spatial domain methods, it can add a large number of watermarks to the image on the premise of the stealthiness of the watermark. Cox et al. proposed a digital watermarking algorithm based on global transformation [43]. It first performs DCT on the image, removes the DC component, and adds the watermark to the low-frequency component. Ruanaidh et al. utilized DFT on the image [44], then added the watermark to the phase information. Kundur et al. proposed a digital watermarking method based on discrete wavelet transform [45]. They embed the watermark on the wavelet coefficients representing the texture and edge information of the image according to the characteristics of human visual and texture masking.

2.3.3 Comparison of digital watermarking and backdoor attacks

There are many similarities between backdoor attacks and digital watermarking. The most obvious one is that both require the embedded content to have good stealthiness and robustness. However, backdoor attacks require that the embedded content is easily learned by the target model and does not interfere with the learning on clean data, so we should reasonably balance the stealthiness and effectiveness of backdoor attacks. Moreover, digital watermarking techniques require that the embedded content be detected from the carrier image, and backdoor attacks do not need to meet this condition. Hence, inspired by digital watermarking techniques, we consider the relationship between effectiveness, stealthiness, and robustness, and propose stealthy and robust attack methods based on frequency domain transform.

3 Methods

In this section, we introduce the proposed attack methods. We start by defining the threat model. Then, we provide the detailed process of our attack methods to illustrate how to mount the attack. Finally, we introduce two backdoor trigger injection algorithms proposed in this paper.

3.1 Threat model

Without loss of generality, we follow the threat model from previous work. It includes:

1. The adversary does not know the relevant information of the training dataset.
2. The adversary does not know the structure and parameter information of the target model.
3. The adversary can inject a small amount of data into the training dataset.

Under this threat model, the adversary cannot directly manipulate the training process and the model, but can add a small amount of data to the training data.

Our attack goals include stealthiness, robustness, and effectiveness. And stealthiness means that the trigger should be natural and not easily detected, and robustness requires that the trigger is still effective after data augmentation operations.

3.2 Attack process

Inspired by digital watermarking techniques, we propose a stealthy and robust backdoor attack. The detailed process is illustrated in Figure 2. It consists of three stages: attack stage, training stage, and inference stage.

3.2.1 Attack stage

During the attack stage, the adversary needs to generate poisoned data containing backdoor triggers. To ensure the stealthiness and robustness of backdoor attacks, we propose two backdoor trigger injection algorithms using frequency domain transform. These algorithms take original data D and predefined trigger t as input, add the trigger t to the frequency domain of the original data, then assign them predefined wrong labels y_t, finally get poisoned data D_t. The specific details of the trigger injection are detailed in Section 3.3.

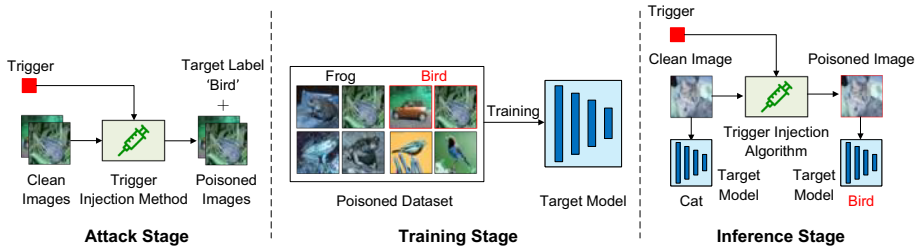


Figure 2 Attack process

3.2.2 Training stage

The adversary merges the generated poisoned data D_t with clean data D_c in the training stage, i.e., $D_p = D_t \cup D_c$. The adversary then uses the poisoned dataset D_p to train the target model $f_\theta(x)$, i.e.,

$$\min_\theta \frac{1}{n} \sum_{(x,y)\in D_p} \mathcal{L}(f_\theta(x), y)$$

where $\mathcal{L}$ is the loss function.

3.2.3 Inference stage

The adversary uses the trigger injection algorithms to add the trigger t to clean data D_c to generate poisoned data D_t, then inputs D_t to the target model, activating the backdoor in the model.

3.3 Trigger injection

To ensure the effectiveness, stealthiness, and robustness of backdoor attacks, we inject backdoor triggers into the frequency domain of the data, and the pipeline is shown in Figure 3.

As shown in Figure 3, we first transform the clean image from the spatial domain to the frequency domain, then inject the trigger into the frequency domain, and finally use the inverse transform to transform the image back to the spatial domain. Specifically,

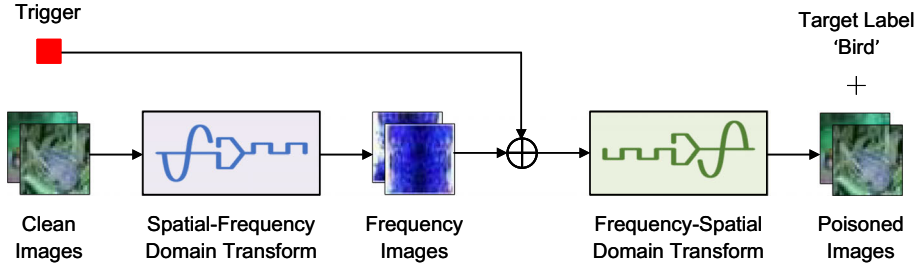


Figure 3 Trigger injection

we design Discrete-Fourier-transform-based and Discrete-cosine-transform-based backdoor trigger injection algorithms.

3.3.1 Discrete-fourier-transform-based backdoor trigger injection algorithm

Mathematically, the Fourier transform represents an image function using a series of periodic functions. Based on the discrete Fourier transform, we propose a backdoor trigger injection algorithm, as shown in Algorithm 1.

Algorithm 1 Discrete-Fourier-transform-based Backdoor Trigger Injection Algorithm (BTI-DFT)

Require: Original Data D, Backdoor Trigger t
Ensure: Poisoned Data D_t
1: $D_f \Leftarrow \mathcal{F}(D)$ ▷ Perform discrete Fourier transform on D
2: $D_f^{i\in\{0,\cdots,N-1\}} \Leftarrow divide(D_f)$ ▷ Divide D_f into N data blocks
3: **for** $j = 0; j < M; j++$ **do** ▷ Select M data blocks
4: $\quad D_{ft}^j \Leftarrow D_f^j + \alpha t$ ▷ Inject the trigger t into D_f^j
5: **end for**
6: $D_t \Leftarrow \mathcal{F}_e(D_{ft})$ ▷ Perform inverse discrete Fourier transform on D_{ft}
7: $D_t \Leftarrow real(D_t)$ ▷ Remove the imaginary part of D_t
8: **return** D_t

Take the original data D and backdoor trigger t as input. Line 1 first transforms the original data D from the spatial domain to the frequency domain using the discrete Fourier transform $\mathcal{F}$ defined as Eq. 1. Next line 2 divides D_f into N data blocks, where *divide* represents the block function. Then the backdoor trigger t is injected into the M data blocks to obtain the D_{ft}^j, as shown in lines 3 to 5. And α represents the intensity coefficient that is used to control the injection intensity of the backdoor trigger. After the trigger is embedded, line 6 uses the inverse discrete Fourier transform $\mathcal{F}_e$ to convert D_{ft} from the frequency domain to the spatial domain. Lines 7 and 8 remove the imaginary part of the data using the $real$ function, and obtain the poisoned data D_t. $\mathcal{F}_e$ is defined as Eq. 2.

$$F(u,v) = \sum_{x=0}^{M-1}\sum_{y=0}^{N-1} f(x,y)e^{-j2\pi(\frac{ux}{M}+\frac{vy}{N})} \tag{1}$$

where $f(x,y)$, $F(u,v)$ are the image, the Fourier transform of the image. M, N are the number of rows and columns of the image.

$$f_t(x,y) = \frac{1}{MN}\sum_{u=0}^{M-1}\sum_{v=0}^{N-1} F_t(u,v)e^{i2\pi(\frac{ux}{M}+\frac{vy}{N})} \tag{2}$$

where $f_t(x,y)$, $F_t(u,v)$ are the poisoned image, the Fourier transform of the poisoned image.

3.3.2 Discrete-cosine-transform-based backdoor trigger injection algorithm

The cosine transform approximates the image using the sum of a set of cosine functions at different frequencies and amplitudes. Mathematically, the cosine transform is the real part of

Algorithm 2 Discrete-cosine-transform-based Backdoor Trigger Injection Algorithm (BTI-DCT)

Require: Original Data D, Backdoor Trigger t
Ensure: Poisoned Data D_t
1: $D^{i\in\{0,\cdots,N-1\}} \Leftarrow divide(D)$ ▷ Divide D into N data blocks
2: **for** $j = 0; j < M; j++$ **do** ▷ Select M data blocks
3: $D_c^j[0] \Leftarrow \mathcal{C}(D^j[0])$ ▷ Perform discrete cosine transform on $D^j[0]$
4: $D_c^j[0] \Leftarrow \alpha t$ ▷ Inject the trigger t into $D_c^j[0]$
5: $D_t^j[0] \Leftarrow \mathcal{C}_e(D_c^j[0])$ ▷ Perform inverse discrete cosine transform on $D_c^j[0]$
6: **end for**
7: **return** D_t

the Fourier transform. Discrete-cosine-transform-based backdoor trigger injection algorithm is shown in Algorithm 2.

In Algorithm 2, lines 1 and 2 first divide the original data D into N data blocks, and select M blocks for trigger injection. Next line 3 selects a channel of the image to perform discrete cosine transform $\mathcal{C}$, which transforms the channel data from the spatial domain to the frequency domain. $\mathcal{C}$ is defined as (3). Then line 4 assigns $D_c^j[0]$ to the value αt. α represents the injection strength coefficient. Finally the inverse discrete cosine transform is used to convert $D_c^j[0]$ from the frequency domain back to the spatial domain, as shown in line 5. $\mathcal{C}_e$ is defined as (4).

$$C(u,v) = c(u)c(v)\sum_{x=0}^{N-1}\sum_{y=0}^{N-1} f(x,y)cos\frac{(2x+1)u\pi}{2N}cos\frac{(2y+1)v\pi}{2N} \tag{3}$$

where $f(x,y), \mathcal{C}(u,v)$ are the image, the cosine transform of the image, $u, v = 0, 1, ..., N-1$, $c(u) = \begin{cases} \sqrt{\frac{1}{N}}, u = 0 \\ \sqrt{\frac{2}{N}}, u \neq 0 \end{cases}$.

$$f_t(x,y) = \sum_{u=0}^{N-1}\sum_{v=0}^{N-1} c(u)c(v)C_t(u,v)cos\frac{(2x+1)u\pi}{2N}cos\frac{(2y+1)v\pi}{2N} \tag{4}$$

where $f_t(x,y), \mathcal{C}_t(u,v)$ are the poisoned image, the cosine transform of the poisoned image, $x, y = 0, 1, ..., N-1, c(u) = \begin{cases} \sqrt{\frac{1}{N}}, u = 0 \\ \sqrt{\frac{2}{N}}, u \neq 0 \end{cases}$.

4 Experiments

In this section, we conduct extensive experiments to evaluate the proposed methods on diverse benchmarking datasets, with comparisons to other representative attack baselines. Specifically, the experiment contents include: (1) Attack Effectiveness; (2) Attack Stealthiness; (3)Attack Robustness; (4) Resistance to Neural Cleanse.

Table 1 The structure of target model

GTSRB		CIFAR10	
Conv1	$[5 \times 5, 1]$	Conv1	$[7 \times 7, 64]$
Max Pool	$[2 \times 2]$	Max Pool	$[3 \times 3]$
Conv2	$[3 \times 3, 1]$	Conv2	$\begin{bmatrix} 3 \times 3, 64 \\ 3 \times 3, 64 \end{bmatrix} \times 2$
Conv3	$[1 \times 1, 1]$	Conv3	$\begin{bmatrix} 3 \times 3, 128 \\ 3 \times 3, 128 \end{bmatrix} \times 2$
FC	350	Conv4	$\begin{bmatrix} 3 \times 3, 256 \\ 3 \times 3, 256 \end{bmatrix} \times 2$
FC	43	Conv5	$\begin{bmatrix} 3 \times 3, 512 \\ 3 \times 3, 512 \end{bmatrix} \times 2$
		Average Pool	$[4 \times 4]$
		FC	10

4.1 Experimental setup

We use the GTSRB and CIFAR10 as benchmark datasets in the following experiments [46, 47]. We choose different neural network structures as target models for different datasets, as shown in Table 1 [48]. For the backdoor triggers, we take a 4×4 white pixel block on the CIFAR10 dataset; since the background color of most data is white, we choose a 4×4 red pixel block on the GTSRB dataset, as shown in Figures 4 and 5. And we set the poisoning ratio to 10% and the strength coefficients α in BTI-DFT and BTI-DCT to 1 and 2. Additionally, we use the FFT module in the Numpy toolkit and the DCT module in the OpenCV toolkit for two-dimensional discrete Fourier transform and two-dimensional discrete cosine transform, respectively. And in all experiments, we compare our methods with two representative existing methods, i.e., Badnets [14] and Blend [17], as the baselines.

4.2 Attack effectiveness

In this experiment, we evaluate the attack effectiveness from the prediction accuracy (CSA) on the clean data and the attack success rate (ASR) on the poisoned data, and the results are shown in Table 2.

From Table 2, on the GTSRB dataset, the ASRs of BTI-DFT and BTI-DCT are 96.97% and 96.67%, and the CSAs are 96.38% and 96.61%. Our methods are slightly weaker than Badnets and Blend. While on the CIFAR10 dataset, BTI-DFT and BTI-DCT are superior to Badnets and Blend in both CSA and ASR. And their ASRs reach 100%. Overall, our methods have good attack performance and can meet the requirements of most real-world scenarios.

4.3 Attack stealthiness

This experiment aims to verify whether the backdoor trigger is easily detected by humans. Specifically, we use Badnets and Blend to inject backdoor triggers in the spatial domain, and use BTI-DFT and BTI-DCT to inject backdoor triggers in the frequency domain. The generated images are shown in Figures 4 and 5.



Figure 4 Trigger injection on GTSRB (Line 1: Clean images; Line 2: Badnets; Line 3: Blend; Line 4: BTI-DFT; Line 5: BTI-DCT)

In Figures 4 and 5, compared with Badnets and Blend, our methods generate more natural poisoned data, and the backdoor triggers are less perceptible. And compared with the original images, BTI-DFT and BTI-DCT only add some irregular and subtle noises. Hence, our methods have better stealthiness and feasibility in real-world scenarios.

4.4 Attack robustness

In this experiment, we verify the robustness of backdoor attacks for the data augmentation operations, i.e., random horizontal flip, random vertical flip, random cropping (padding=2), and random cropping (padding=4), and the results are shown in Figures 6 and 7.

From Figures 6 and 7, compared with the Badnets and Blend, our methods improve at least 80.99%, 68.09%, and 63.31% for random horizontal flip, random vertical flip, and random cropping (padding=4) on the GTSRB dataset. The improvement on random cropping (padding=2) is relatively small, about 25.49%∼56.06%. On the CIFAR10 dataset, compared with the Badnets and Blend, BTI-DFT has a large improvement in random horizontal flip

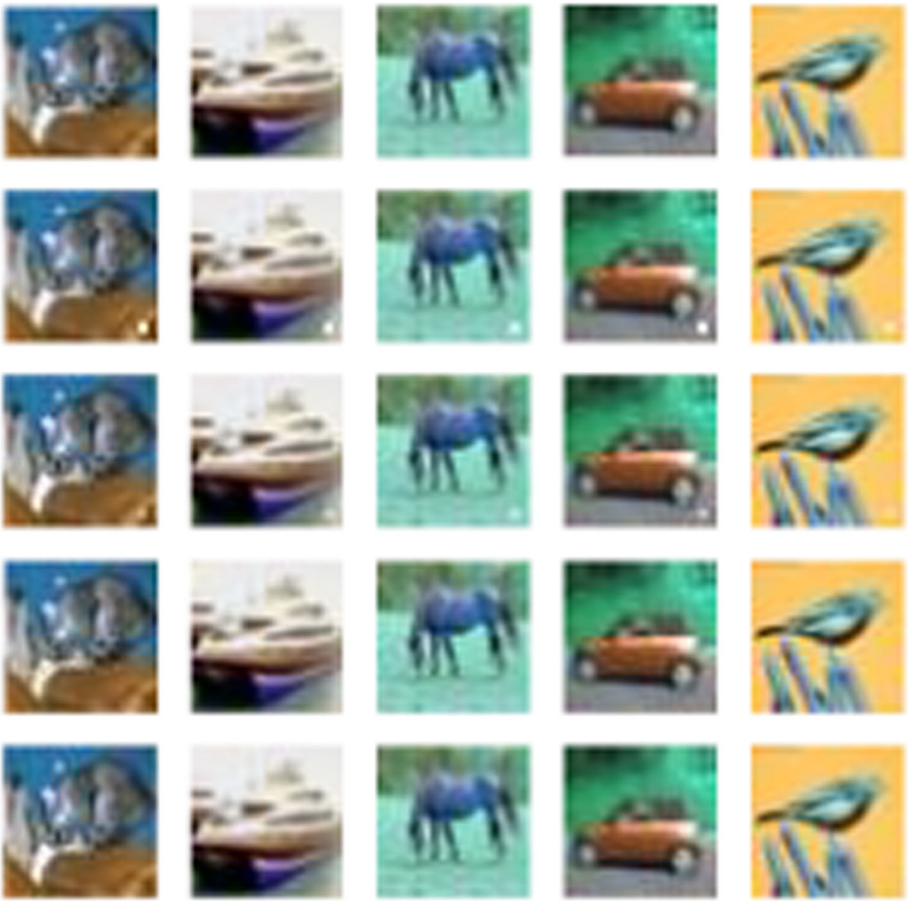


Figure 5 Trigger injection on CIFAR10 (Line 1: Clean images; Line 2: Badnets; Line 3: Blend; Line 4: BTI-DFT; Line 5: BTI-DCT)

and random vertical flip, about 33.91% and 101.15%, while the improvement of BTI-DCT is slight. On the contrary, BTI-DCT has a great improvement in random cropping (padding=2) and random cropping (padding=4), about 268.58% and 157.61%, and BTI-DFT is improved by 67.37% and 83.06%. Hence, our methods are more robust to data augmentation than Badnets and Blend. And BTI-DFT is slightly stronger than BTI-DCT.

Table 2 The performance comparison of attack methods

Attack Methods	GTSRB		CIFAR10	
	CSA(%)	ASR(%)	CSA(%)	ASR(%)
Badnets	**97.62**	**100**	88.02	97.86
Blend	97.47	99.99	87.8	96.82
BTI-DFT	96.38	96.97	**89.04**	**100**
BTI-DCT	96.61	96.67	88.2	**100**

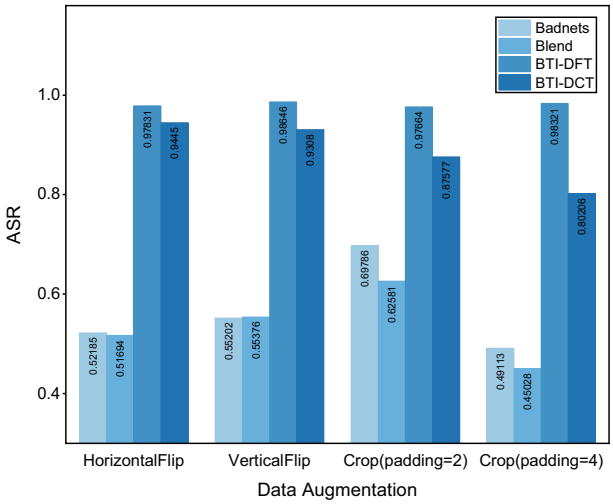


Figure 6 The robustness of backdoor attack on GTSRB

4.5 Resistance to neural cleanse

Neural Cleanse is a classical backdoor detection method that could calculate the target model's anomaly index and determine the backdoor class [32]. In this experiment, we verify the resistance abilities of attack methods to Neural Cleanse, and the results are shown in Figures 8 and 9.

As shown in Figures 8 and 9, on the GTSRB dataset, the anomaly index of BTI-DFT is smaller than that of Badnets and higher than that of Blend; the anomaly index of BTI-DCT is significantly smaller than that of Badnets and Blend. On the CIFAR10 dataset, both BTI-

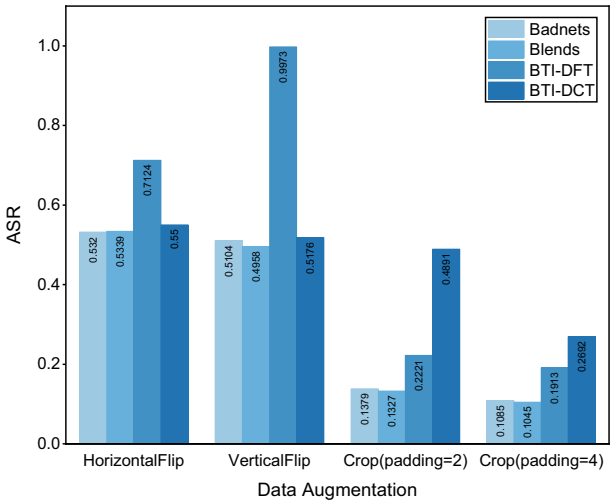


Figure 7 The robustness of backdoor attack on CIFAR10

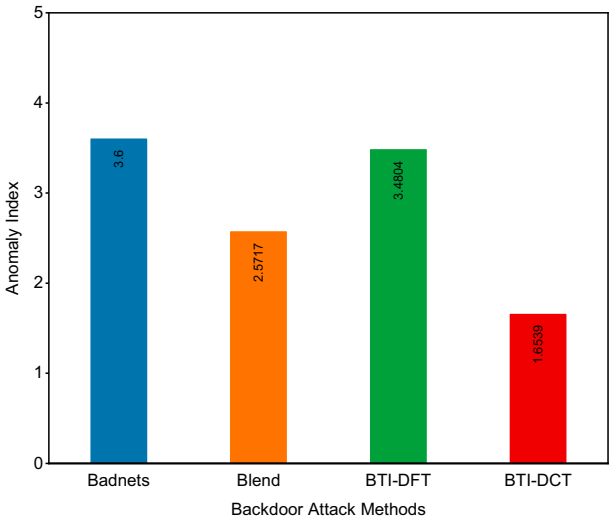


Figure 8 Anomaly index on GTSRB

DFT and BTI-DCT have significantly smaller anomaly indices than Badnets and Blend. This indicates that our methods can make the victim model exhibit lighter anomalies and be more challenging to detect than the Badnets and Blend.

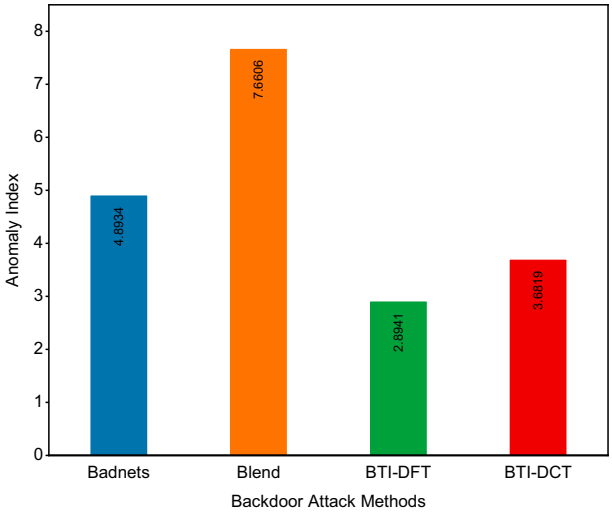


Figure 9 Anomaly index on CIFAR10

5 Conclusion

Inspired by digital watermarking techniques, we propose stealthy and robust backdoor attack methods that meet the requirements of real-world scenarios. Specifically, we design two backdoor trigger injection algorithms using DFT and DCT. These algorithms transform the image from the spatial domain to the frequency domain to the frequency domain, then inject backdoor triggers into the frequency domain instead of the spatial domain to ensure the stealthiness. And they inject triggers multiple times, improving the robustness. We experimentally evaluated the proposed methods, and the results indicate that our methods make backdoor triggers more natural and less perceptible, and these triggers could resist various data augmentation operations, such as random flip, random crop, etc. Moreover, our methods could make the victim models exhibit lighter anomalies, which further increases the stealthiness of backdoor attacks.

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Availability of data and material The datasets used or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval Not applicable for both human and/or animal studies.

Competing interests The authors declare that they have no competing interests.

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